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Ministry of Higher Education and Scientific Research

National Higher School of Advanced Technologies

وزارة التعليم العالي والبحث العلمي

المدرسة الوطنية العليا للتكنولوجيات المتقدمة

Worker internship carried out at

SONATRACH /TRC-Région Transport Centre

Student in 2nd year of Second Cycle Engineering

Telecommunication and Networking systems

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Grade :

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THANK YOU

I would like to extend my heartfelt gratitude to the manager and staff of SONATRACH for their invaluable support and guidance throughout my internship. I am sincerely thankful for their generosity in sharing their time, knowledge, and experiences with me. Their mentorship has been instrumental in enriching my understanding of the industry and enhancing my professional growth. I am truly privileged to have had the opportunity to learn from such dedicated and talented individuals. Thank you for making my internship experience at SONATRACH both rewarding and unforgettable.

Introduction :

The internship took place at SONATRACH, the national oil and gas company, renowned for its expertise in hydrocarbon exploration, production, transportation, refining, and marketing. This provided an opportunity to gain insights into various telecommunication equipment related SCADA (Supervisory Control and Data Acquisition) systems. This internship provided a deep understanding of computer networking within a skilled team, enhancing my self-learning capabilities in emerging technologies and furthering my knowledge acquired during the Engineering program. Additionally, it emphasized the importance of teamwork in a professional environment.

I. Company Description :

SONATRACH, Algeria's state-owned oil and gas company, founded in 1963, is a key player in Africa's and the world's energy market. It engages in all aspects of the industry, from exploration and production to refining and marketing. With its headquarters in Algiers, SONATRACH operates extensively across Algeria, managing oil and gas fields, pipelines, refineries, and distribution facilities. It also plays a significant role internationally, supplying natural gas to Europe and beyond, and engaging in activities such as pipeline transportation, petroleum product marketing, and maritime transport. Through its commitment to innovation and sustainable development, SONATRACH contributes to Algeria's energy sector and global energy landscape.

Pipeline Transport (TRC, Transport par canalisation) [1]

At its creation on 31 December 1963, SONATRACH set itself the task of transport and marketing of hydrocarbons extracted from Sahara deposits by the first foreign companies operating in Algeria at the time. The first project was the OZ1 pipeline linking Haoud-El-Hamra to Arzew, in 1966. The Pipeline Transportation activity is grouped into divisions:

- Operations Division.
- Maintenance Division.

It is provided by five transport regions

- •Ain-Amenas RTI Transport Region.
- • Haoud El Hamra RTH Transportation Region.
- • Bejaia RTC Transport Centre Region.
- • Arzew RTO Transportation West Region.
- • SKIKDA RTE East Transportation Region.

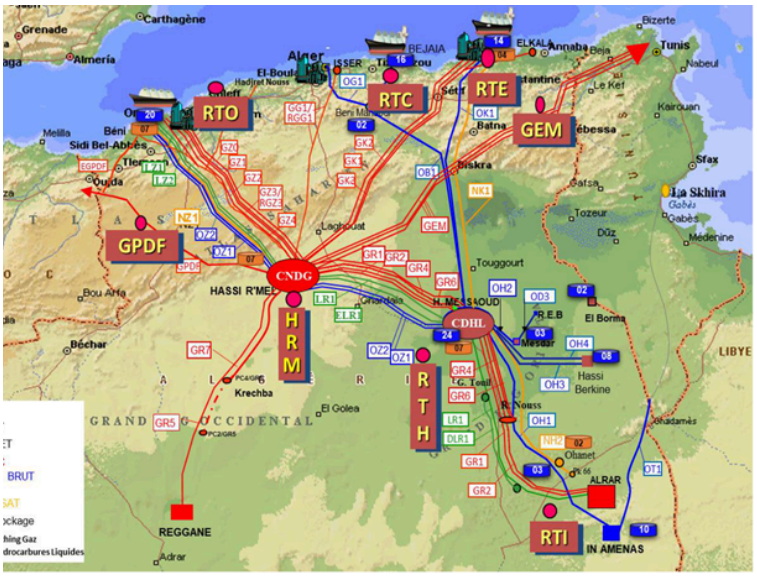

Figure 1: *Sonatrach pipeline transport network.*

Organisation de la DRGB :

It is organized according to the organization chart in the following figure:

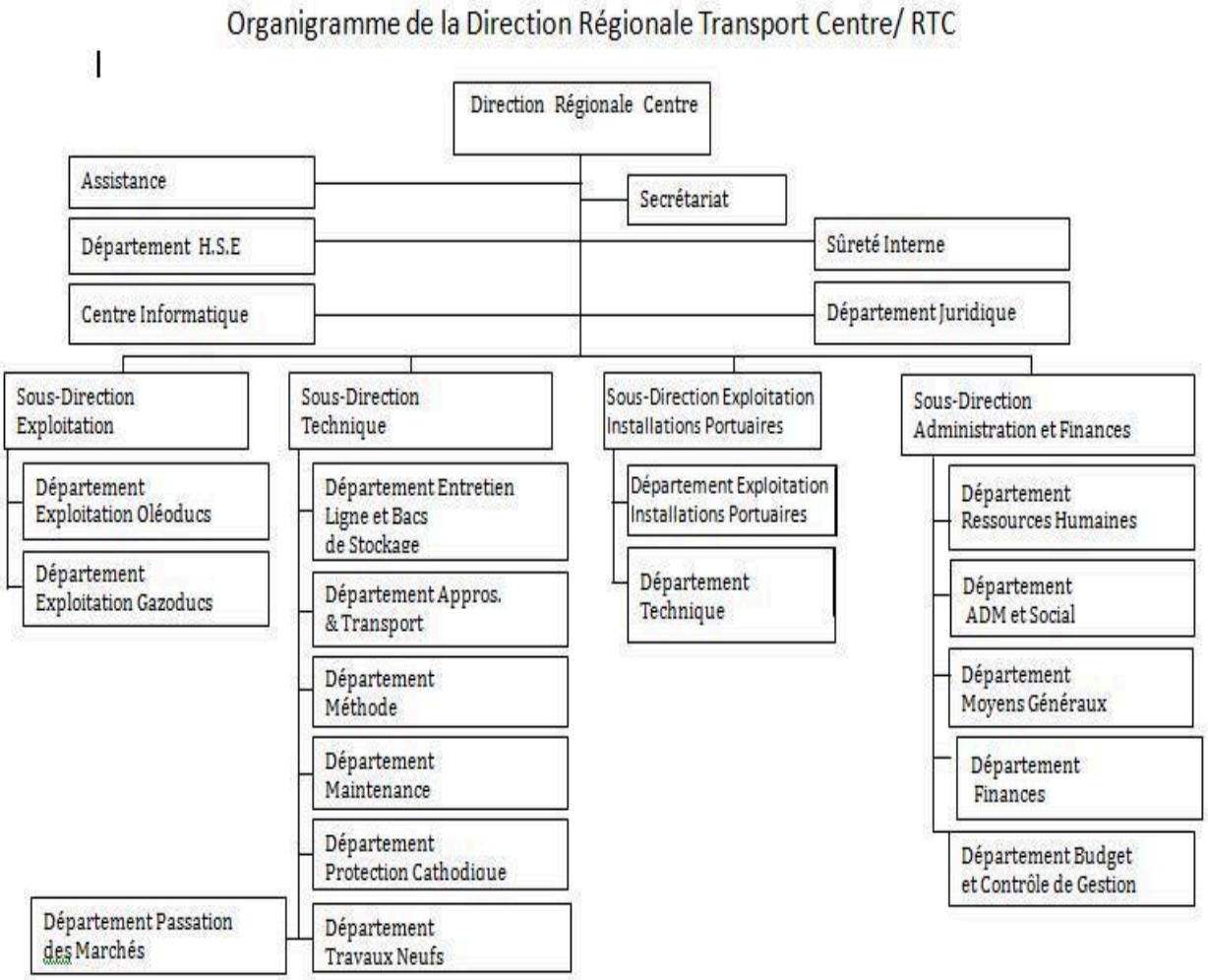

Figure 2 : *Organizational chart of the Regional Transport Center/RTC Directorate.*

The maintenance Department [1]

This very important department within the DRGB, It includes maintenance services of the hydrocarbon transport line, it is at the forefront in relation to the importance of these activities. This department administers the management of services provided by operational branches such as functional departments which develop and ensure the application of a group policy and strategy. They provide the expertise and support necessary for activities. It is divided into five services, its structure is shown in the following figure :

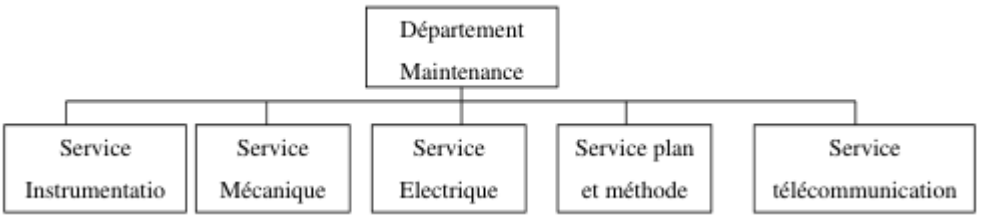

Figure 3: *Maintenance Department*

II. The Internship within the company :

SCADA System [2][3]

It is a remote supervision, remote transmission and telemetry system, it translates the information transmitted by the pipe into the form of data (temperature, flow rate, etc.) which is routed to the incoming terminal where it is translated by a controller . All communications are managed from the auto-com room, which contains cabinets with frames specially made to install all the elements that allow communications to be properly managed: the processor, internal network card, CPU card, signaling card (MMT), patch panel (optical fiber)…etc.

The controle Tour [2][3]

The control room or command room is used to control and monitor the actions generated around any entity. It is equipped with two synoptic tables, one used for the buoy control systems and the other to visualize all the filling and emptying operations of the bins at the oil port of the two terminals. The control operations are: Controls line pressure and flow. Control of filling and emptying of bins with automatic display. Valve and pump controls. Pressure control in the loading line. Oil loading control (crude oil) to the ship.

Figure 4: *Control and Supervision room*

SCADA Architecture :

SCADA architecture refers to its structure and components. This architecture is generally illustrated in Figure 5. From a software point of view, we can speak of a layered architecture, comprising a client layer, processing layer, and data [4].

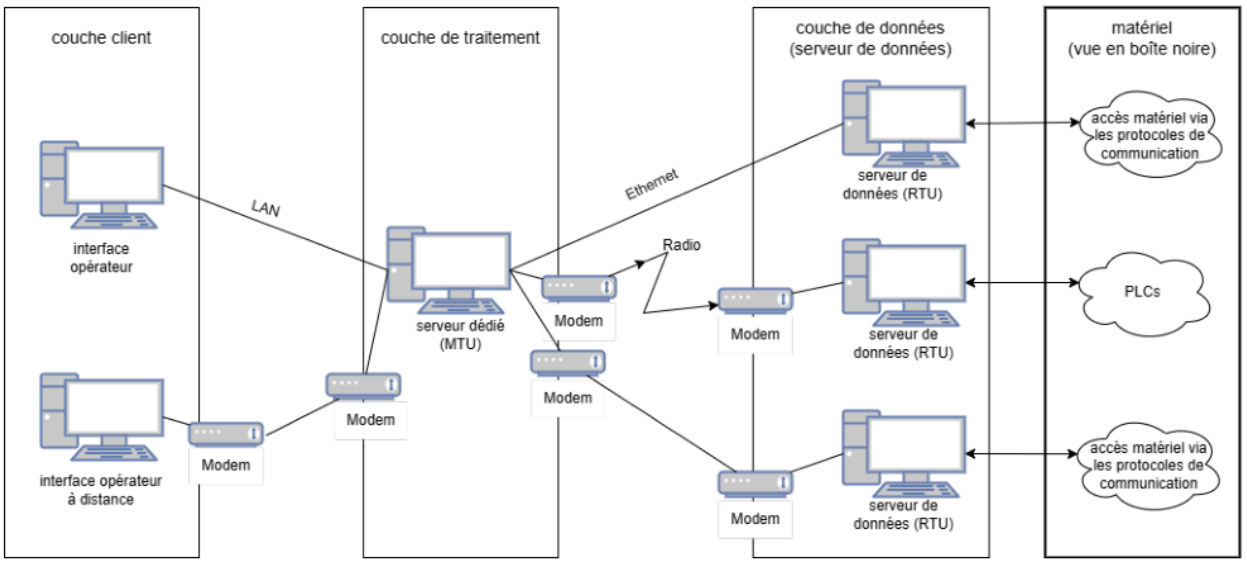

Figure 5: *General architecture of SCADA Systems*

This architecture illustrates the following aspects:

− The separation of client, processing and data layers.

− The distinction between SCADA components and controlled equipment.

− The different means of communication used for the exchange of data between the client layer and the processing layer, between the processing layer and the data, and finally between the data layer and the hardware [5].

Communication Support :

SCADA systems can use different types of networks for transmission of data, such as Ethernet networks, serial networks (RS-232, RS-485), cellular networks (GSM, GPRS, LTE), wireless networks, etc. The choice of network depends on specific system requirements, communication distance, required bandwidth and other factors .

Architecture client-serveur

In SCADA systems, client-server architecture is commonly used for communication. Mainframes or SCADA servers act as nodes central, receiving data from sensors, sending commands to actuators and processing information for display and analysis. User interfaces, stations monitoring and remote terminals act as clients that interact with the SCADA server.

Data acquisition SCADA systems collect data from various monitoring points such as sensors, meters and programmable controllers. These data are generally grouped into logical blocks and collected periodically or based on events specific. Data collected may include real-time measurements, conditions alarms, events, diagnostic reports, etc.

Data transfer Data acquired in SCADA systems is transferred via networks of communication to the different nodes of the system. Data can be transmitted synchronously or asynchronously, in real time or in batch mode. Data transfer can be based on periodic queries, event notifications or requests specific to the system .

Communications security SCADA systems must have security measures to protect communications against intrusions, malicious attacks or interference. That includes the use of authentication mechanisms, data encryption, firewalls, network segmentation, secure protocols, etc.

Conclusion :

In conclusion, my internship at SONATRACH has been a transformative experience, providing invaluable insights into the dynamic field of telecommunications within the context of a leading oil and gas company. Through hands-on involvement with VoIP and SCADA systems, I gained practical skills that complimented my academic knowledge and furthered my professional development. Working alongside a competent team at SONATRACH not only deepened my understanding of computer networking but also reinforced the importance of collaboration and teamwork in achieving organizational objectives. This internship has not only enriched my technical capabilities but has also instilled in me a greater appreciation for the complexities of the energy industry and its reliance on advanced telecommunications infrastructure. I am grateful for the opportunity to contribute to SONATRACH's mission and am confident that the experiences gained during this internship will serve as a solid foundation for my future endeavors in the field of engineering.

References :

[1] http://www.sonatrach.com.
[2] S. A.Boyer, «definition of scada,» chez scada:supervisory control and data acquisition, 2004, p. 9.
[3] S. A.Boyer, «What is SCADA?,» chez SCADA:Supervisory Control And Data Acquisition , 2004, p. 9.
[4] L. S. Jan Stowisek, «General Architecture of SCADA Systems,» A PVSS Application for Monitoring the Start-up of the Super Proton Synchrotron after Major Breakdowns, pp. 15-16-17, 2001.
[5] A. M. MEZDOUR Hala, «Etude et réalisation d'un système de supervision sous,» 2018/2019.